2D Vectors

Questions 1–44

Question 1. It is given that A and B have position vectors $\begin{pmatrix}2\\6\end{pmatrix}$ and $\begin{pmatrix}-3\\8\end{pmatrix}$ respectively.

a) Find $\overrightarrow{AB}$.

b) Find $|\overrightarrow{AB}|$.

c) Find a unit vector in the direction of $\overrightarrow{AB}$.

Question 2. It is given that P and Q have position vectors $4\mathbf{i}-\mathbf{j}$ and $7\mathbf{i}+8\mathbf{j}$ respectively. Find a vector of length $6\sqrt{10}$ in the direction of $\overrightarrow{PQ}$.

Question 3. It is given that S and T have position vectors $\begin{pmatrix}\sqrt{150}\\ \sqrt{162}\end{pmatrix}$ and $\begin{pmatrix}\sqrt{96}\\ \sqrt{98}\end{pmatrix}$ respectively. Find a vector of length $\sqrt{126}$ in the direction of $\overrightarrow{ST}$.

Question 4. It is given that A and B have position vectors $(k-6)\mathbf{i}+(2k+4)\mathbf{j}$ and $(2k-3)\mathbf{i}+(3k+1)\mathbf{j}$ respectively, where k is a constant.

a) The vector $\mathbf{p}$ is of length $\sqrt{4k^2+36}$ in the direction of $\overrightarrow{AB}$. Find the vector $\mathbf{p}$.

b) Given that k is a positive integer, find the shortest possible vector $\mathbf{p}$.

Question 5. Find the scalar product of

a) $\mathbf{a} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$

b) $\mathbf{a} = (4\mathbf{i} + \mathbf{j})$ and $\mathbf{b} = (-2\mathbf{i} + 5\mathbf{j})$

c) $\mathbf{a} = \begin{pmatrix} 0 \\ -4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -3 \\ -6 \end{pmatrix}$

d) $\mathbf{a} = 9\mathbf{j}$ and $\mathbf{b} = 6\mathbf{i} - \mathbf{j}$

Question 6. Find the value of $(3p\mathbf{i} - 2\mathbf{j}) \cdot (-5\mathbf{i} - 4p\mathbf{j})$ in terms of p.

Question 7. Given that q is nonzero, find the value of $\left(8q^3\mathbf{i} + \frac{5}{q}\mathbf{j}\right) \cdot \left(-\frac{3}{q^2}\mathbf{i} + 9q^2\mathbf{j}\right)$ in terms of q.

Question 8. Use the scalar product to find the angle between the following pairs of vectors. Give your answer in degrees to 1 decimal place.

a) $\mathbf{a} = \begin{pmatrix} 1 \\ 9 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$

b) $\mathbf{a} = -2\mathbf{i} + 10\mathbf{j}$ and $\mathbf{b} = 3\mathbf{i} + 9\mathbf{j}$

c) $\mathbf{a} = \begin{pmatrix} -2 \\ -11 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 10 \\ 3 \end{pmatrix}$

d) $\mathbf{a} = -8\mathbf{i} + \mathbf{j}$ and $\mathbf{b} = -10\mathbf{i} + 2\mathbf{j}$

Question 9. It is given that $A(2,4)$, $B(-2,1)$ and $C(-1,5)$. Find the angle ABC to 1 decimal place.

Question 10. Given that the vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular, state the value of $\mathbf{a} \cdot \mathbf{b}$.

Question 11. In each case, determine if the following pairs of vectors are perpendicular.

a) $\mathbf{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ -3 \end{pmatrix}$

b) $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -2\mathbf{i} + 5\mathbf{j}$

c) $\mathbf{a} = \begin{pmatrix} -8 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 8 \end{pmatrix}$

d) $\mathbf{a} = \sqrt{8}\mathbf{i} - \sqrt{32}\mathbf{j}$ and $\mathbf{b} = \sqrt{72}\mathbf{i} + \sqrt{18}\mathbf{j}$

Question 12. Determine the value(s) of p that would make the following pairs of vectors perpendicular:

a) $\mathbf{a} = \begin{pmatrix} 2p \\ 4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5 \\ -3 \end{pmatrix}$

b) $\mathbf{a} = 6\mathbf{i} - 4p\mathbf{j}$ and $\mathbf{b} = 8p\mathbf{i} + \frac{1}{2}\mathbf{j}$

c) $\mathbf{a} = \begin{pmatrix} 4+p \\ 3p \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2p \\ 1-p \end{pmatrix}$

d) $\mathbf{a} = (2p-1)\mathbf{i} + (3p-1)\mathbf{j}$ and $\mathbf{b} = (p-2)\mathbf{i} + (2p+3)\mathbf{j}$

Question 13. Given that k is a constant, show that the vectors $\mathbf{a} = \begin{pmatrix} 3k+2 \\ 2k+2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -6k-6 \\ 9k+6 \end{pmatrix}$ are perpendicular for all values of k.

Question 14. Given that k is a constant, show that the vectors $\mathbf{a} = \begin{pmatrix} k+1 \\ 6k \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2-k \\ 4k+1 \end{pmatrix}$ can never be perpendicular.

Question 15. Find the vector equation of the line that:

a) passes through $(3, 1)$ and is parallel to $2\mathbf{i} + \mathbf{j}$

b) passes through $(-2, -5)$ and is parallel to $-\mathbf{i} + 9\mathbf{j}$

c) passes through $(0, -8)$ and is parallel to $\mathbf{j}$

d) passes through (a, b) and is parallel to $p\mathbf{i} + q\mathbf{j}$, where a, b, p and q are constants

Question 16. Find a vector equation of the line that:

a) passes through $(1, 4)$ and $(2, 9)$

b) passes through $(3, 5)$ and $(5, 15)$

c) passes through $(-2, 1)$ and $(0, -5)$

d) passes through $(-4, 3)$ and $(4, -13)$

e) passes through $(2, 9)$ and $(-3, -20)$

f) passes through $(-4, -1)$ and $(-10, -7)$

g) passes through $(k, k + 1)$ and $(k - 2, 2k)$, where k is a constant

Question 17. Determine whether the point $(3, 2)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} 5 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

Question 18. Determine whether the point $(3, -2)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} -4 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

Question 19. Determine whether the point $(-15, 15)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} -7 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \end{pmatrix}$.

Question 20. It is given that the point $(4, -2)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} -3 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} k \\ -2 \end{pmatrix}$, where k is a constant. Find the value of k.

Question 21. The point $(-3, -8)$ lies on the line with equation $\mathbf{r} = \begin{pmatrix} 4 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ k \end{pmatrix}$, where k is a constant. Find the value of k.

Question 22. Determine whether the following pairs of lines are coincident, parallel or if they intersect. If they intersect, find the coordinates of the point of intersection:

a) $l_1 : \mathbf{r} = \begin{pmatrix} -2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -5 \\ 2 \end{pmatrix}$

b) $l_1 : \mathbf{r} = \begin{pmatrix} 4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} -5 \\ 21 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ 5 \end{pmatrix}$

c) $l_1 : \mathbf{r} = \begin{pmatrix} 6 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 9 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 25 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ -18 \end{pmatrix}$

d) $l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -6 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 6 \end{pmatrix}$

Question 23. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 3 \\ 10 \end{pmatrix} + \lambda \begin{pmatrix} -12 \\ 3 \end{pmatrix}$, and the line l_2 is parallel to the vector $5\mathbf{i} + 20\mathbf{j}$. Show that l_1 and l_2 are perpendicular.

Question 24. Determine whether the following pairs of lines are perpendicular or not:

a) $l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -5 \\ 3 \end{pmatrix}$

b) $l_1 : \mathbf{r} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -7 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ -7 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 7 \end{pmatrix}$

c) $l_1 : \mathbf{r} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 9 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} -9 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 9 \end{pmatrix}$

d) $l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ -11 \end{pmatrix} + \lambda \begin{pmatrix} 11 \\ 3 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 9 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 11 \end{pmatrix}$

Question 25. It is given that the following lines are perpendicular:

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ -100 \end{pmatrix} + \lambda \begin{pmatrix} k \\ -3 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 2 \\ 557 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 9 \end{pmatrix},$$

where k is a constant.

a) Find the value of k.

b) Find the position vector of the point of intersection of the two lines.

Question 26. It is given that the following lines are perpendicular:

$$l_1 : \mathbf{r} = \begin{pmatrix} 4 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ k \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} -2 \\ 14 \end{pmatrix} + \mu \begin{pmatrix} 20 \\ 20 \end{pmatrix},$$

where k is a constant.

a) Find the value of k.

b) l_1 and l_2 intersect at the point P. Find the coordinates of the point P.

c) Find a unit vector in the direction of the position vector of P.

Question 27. It is given that the following lines intersect and are not coincident:

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 15 \\ k \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} -3 \\ -4 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 2 \end{pmatrix},$$

where k is a constant. Find the possible values of k.

Question 28. It is given that

$$l_1 : \mathbf{r} = \begin{pmatrix} 5 \\ 420 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -10 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 4 \\ 120 \end{pmatrix} + \mu \begin{pmatrix} 10 \\ -3 \end{pmatrix}.$$

a) Find the coordinates of the point of intersection of the two lines.
b) It is given that the line l_3 passes through the point $(90, 100)$ and the intersection of l_1 and l_2. Find the equation of the line l_3.

Question 29. Find the acute angle between each pair of lines:

a) $l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 9 \end{pmatrix}$

b) $l_1 : \mathbf{r} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 2 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -5 \\ 11 \end{pmatrix}$

c) $l_1 : \mathbf{r} = \begin{pmatrix} -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 1 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

d) $l_1 : \mathbf{r} = \begin{pmatrix} -6 \\ 13 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ -2 \end{pmatrix}$

Question 30. Find the acute angle between the line $\mathbf{r} = \begin{pmatrix} 3 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -5 \end{pmatrix}$ and the:

a) x-axis.

b) y-axis.

Question 31. Find the obtuse angle between the following pair of lines: $l_1 : \mathbf{r} = 3\mathbf{i} - 2\mathbf{j} + \lambda(4\mathbf{i} - \mathbf{j})$, $l_2 : \mathbf{r} = 6\mathbf{i} + \mathbf{j} + \mu(3\mathbf{i} + 4\mathbf{j})$.

Question 32. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \end{pmatrix}$. l_1 is mapped to its image l_2 by the matrix $\begin{pmatrix} 3 & 2 \\ -1 & 5 \end{pmatrix}$. Find the equation of the line l_2.

Question 33. The line l_1 has equation $\mathbf{r} = 6\mathbf{i} - 3\mathbf{j} + \lambda(2\mathbf{i} - 7\mathbf{j})$. l_1 is mapped to its image l_2 by the matrix $\begin{pmatrix} 4 & 1 \\ 2 & -3 \end{pmatrix}$. Find the equation of the line l_2.

Question 34. The line l_1 has equation $\mathbf{r} = \begin{pmatrix} -9 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -8 \end{pmatrix}$. l_1 is mapped to its image l_2 by each of the following transformations. Find the equation of the line l_2 in each case:

a) Stretch, parallel to the x-axis, scale factor 5
b) Enlargement, centred at 0, scale factor 2
c) Reflection in the x-axis
d) Reflection in the y-axis
e) Rotation, anticlockwise about the origin, by 90°
f) Rotation, anticlockwise about the origin, by 180°

Question 35. Find the shortest distance from the origin to the line with equation $y = 2x + 5$.

Question 36. Find the shortest distance from the point $(3, -5)$ to the line with the equation $y = 8x + 1$.

Question 37. Find the shortest distance from the point $(4, -2)$ to the line with equation $\mathbf{r} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

Question 38. The shortest distance from the point $(1, k)$ to the line with equation $\mathbf{r} = \begin{pmatrix} 4 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 3 \end{pmatrix}$ is $14\sqrt{34}$ units. Find the integer value of k.

Question 39. Find the shortest distance between the following pair of parallel lines: $l_1 : y = x + 2$, $l_2 : y = x + 8$.

Question 40. Find the shortest distance between the following pair of parallel lines: $l_1 : y = 5x - 20$, $l_2 : y = 5x + 30$.

Question 41. Find the shortest distance between the following pair of parallel lines: $l_1 : y = 2 - 10x$, $l_2 : y = -1 - 10x$.

Question 42. Find the shortest distance between the following pair of parallel lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

Question 43. Find the shortest distance between the following pair of parallel lines:

$$l_1 : \mathbf{r} = \begin{pmatrix} 2 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ 5 \end{pmatrix}, \quad l_2 : \mathbf{r} = \begin{pmatrix} -3 \\ -4 \end{pmatrix} + \mu \begin{pmatrix} -4 \\ 5 \end{pmatrix}.$$

Question 44. It is given that $l_1 : \mathbf{r} = \begin{pmatrix} -4 \\ k \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ and $l_2 : \mathbf{r} = \begin{pmatrix} 8 \\ 9 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 5 \end{pmatrix}$ are parallel. It is also given that the shortest distance between l_1 and l_2 is $4\sqrt{29}$ units. Find the possible values of k.